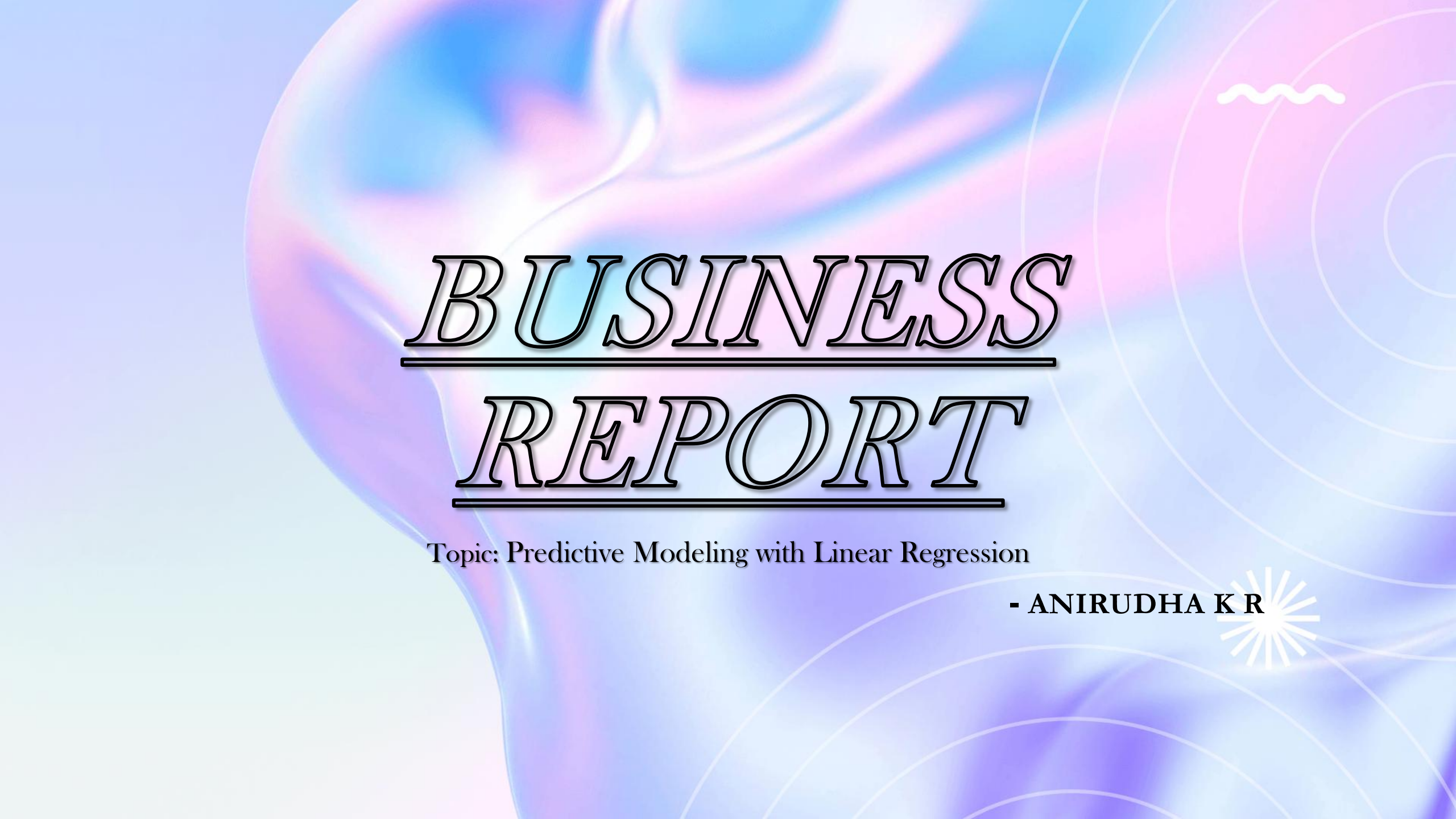




BUSINESS *REPORT*

Topic: Predictive Modeling with Linear Regression

- ANIRUDHA K R

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1. Context

An over-the-top (OTT) media service is a media service offered directly to viewers via the internet. The term is most synonymous with subscription-based video-on-demand services that offer access to film and television content, including existing series acquired from other producers, as well as original content produced specifically for the service. They are typically accessed via websites on personal computers, apps on smartphones and tablets, or televisions with integrated Smart TV platforms.

Presently, OTT services are at a relatively nascent stage and are widely accepted as a trending technology across the globe. With the increasing change in customers' social behavior, which is shifting from traditional subscriptions to broadcasting services and OTT on-demand video and music subscriptions every year, OTT streaming is expected to grow at a very fast pace. The global OTT market size was valued at \$121.61 billion in 2019 and is projected to reach \$1,039.03 billion by 2027, growing at a CAGR of 29.4% from 2020 to 2027. The shift from television to OTT services for entertainment is driven by benefits such as on-demand services, ease of access, and access to better networks and digital connectivity.

With the outbreak of COVID-19, OTT services are striving to meet the growing entertainment appetite of viewers, with some platforms already experiencing a 46% increase in consumption and subscriber count as viewers seek fresh content. With innovations and advanced transformations, which will enable the customers to access everything they want in a single space, OTT platforms across the world are expected to increasingly attract subscribers on a concurrent basis.

2. Exploratory Data Analysis

❖ Problem:

ShowTime is an OTT service provider and offers a wide variety of content (movies, web shows, etc.) for its users. They want to determine the driver variables for first-day content viewership so that they can take necessary measures to improve the viewership of the content on their platform. Some of the reasons for the decline in viewership of content would be the decline in the number of people coming to the platform, decreased marketing spend, content timing clashes, weekends and holidays, etc.

- ShowTime wants to identify the important factors affecting first-day content viewership on its OTT platform.
- **Target Variable:** Views Content
- **Business Goal:** Build insights and identify driver variables influencing content performance.

❖ Data Background:

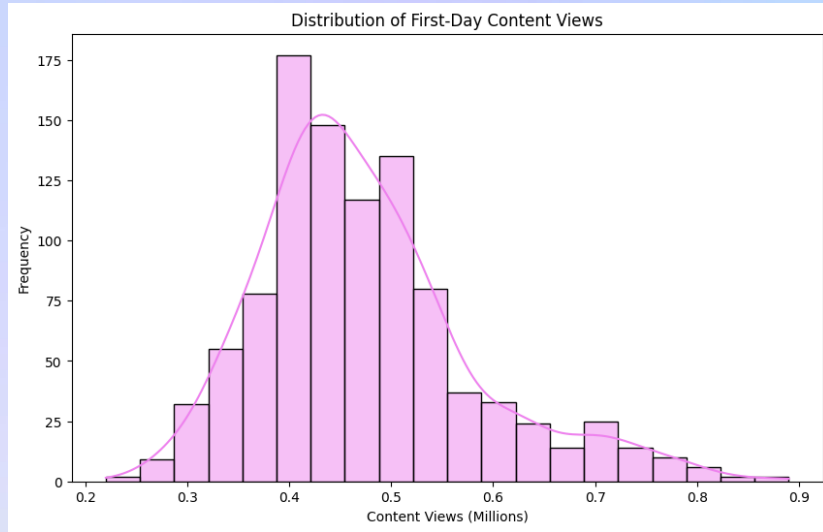
➤ **Index:**

- Visitors
- AD Impressions
- Major Sports Event
- Genre
- Day of week
- Season
- Views Trailer
- Views Content

❖ Univariate analysis

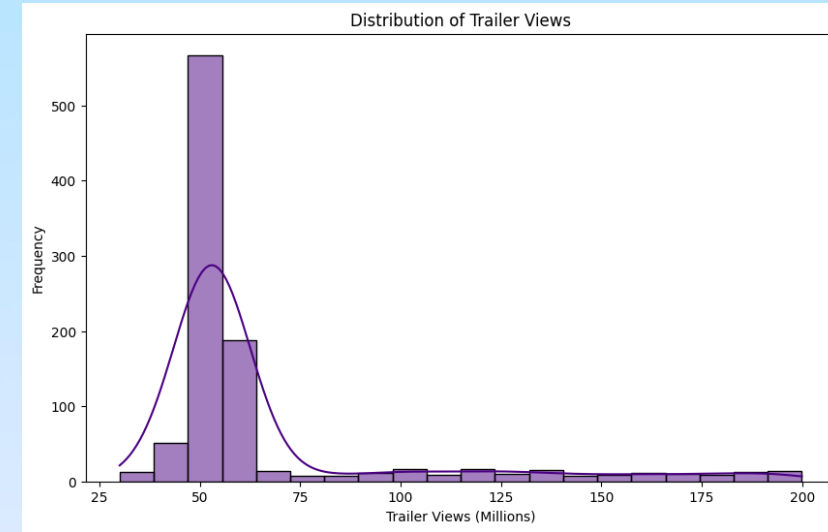
➤ Distribution of content views

Insight: Most content receives moderate first-day viewership.

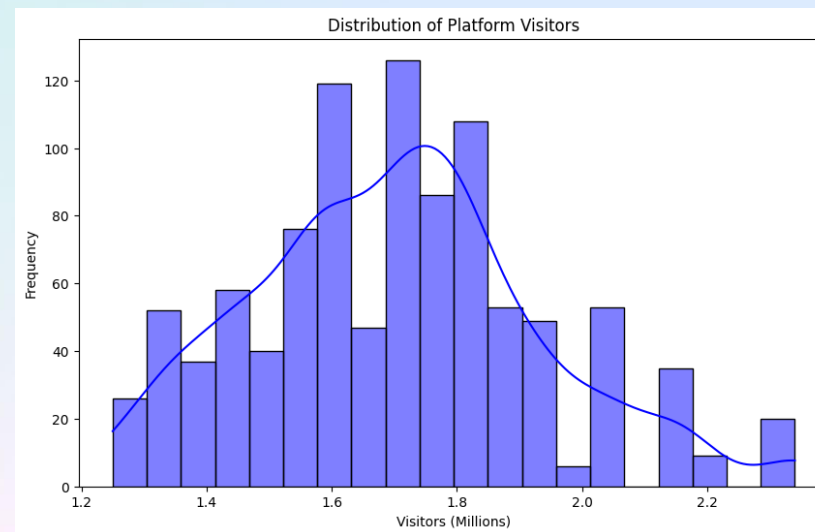


➤ Distribution of trailer views

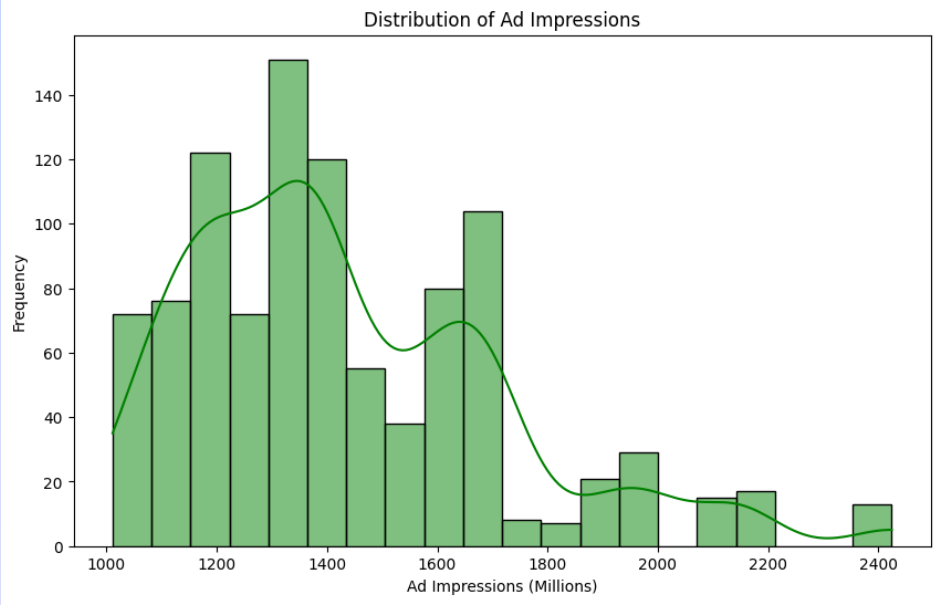
Insight: Trailer views vary significantly across contents.



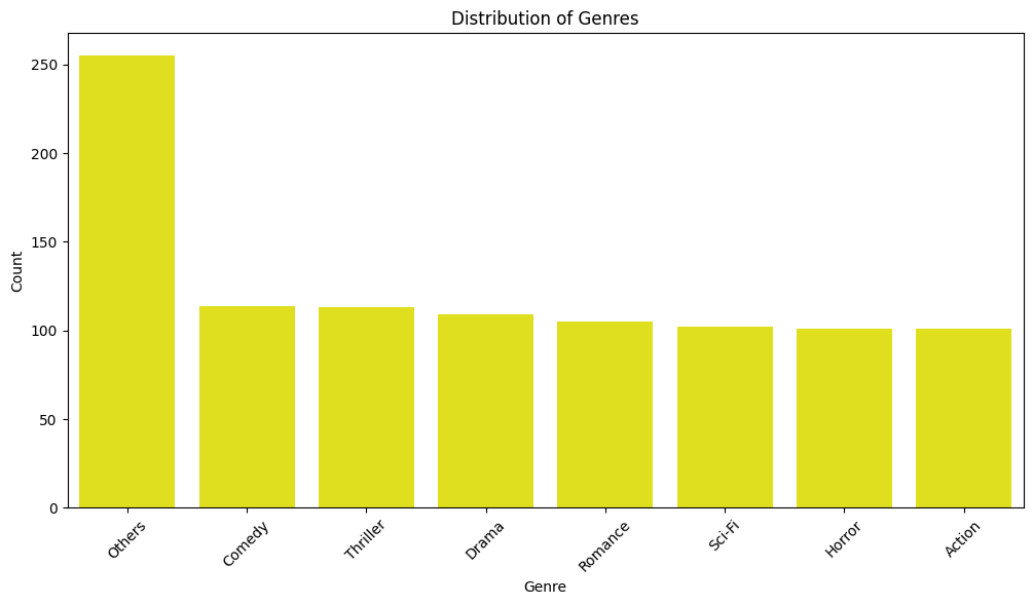
➤ Distribution of visitors



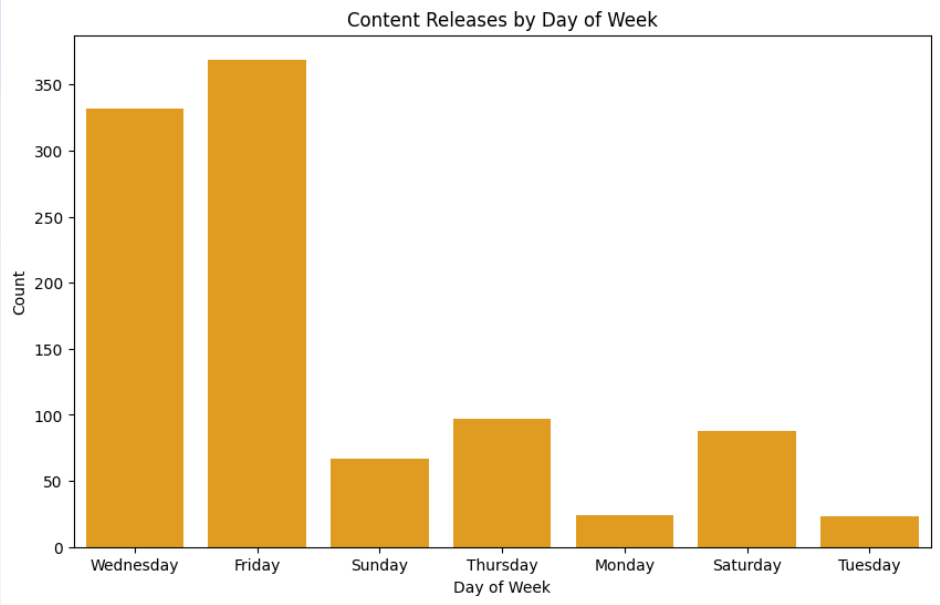
➤ Distribution of impressions



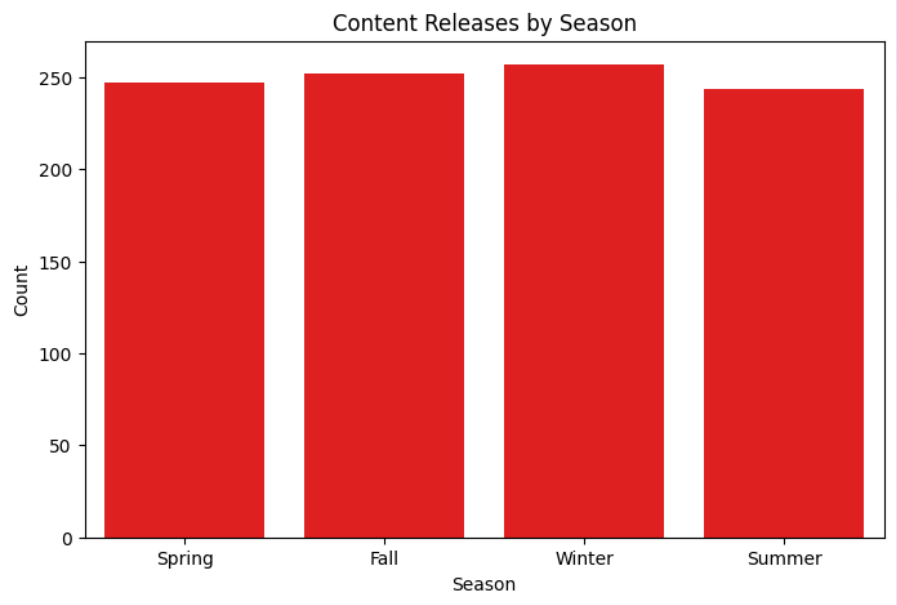
➤ Genre distribution (Some genres dominate the OTT platform content library.)



➤ Day of week distribution

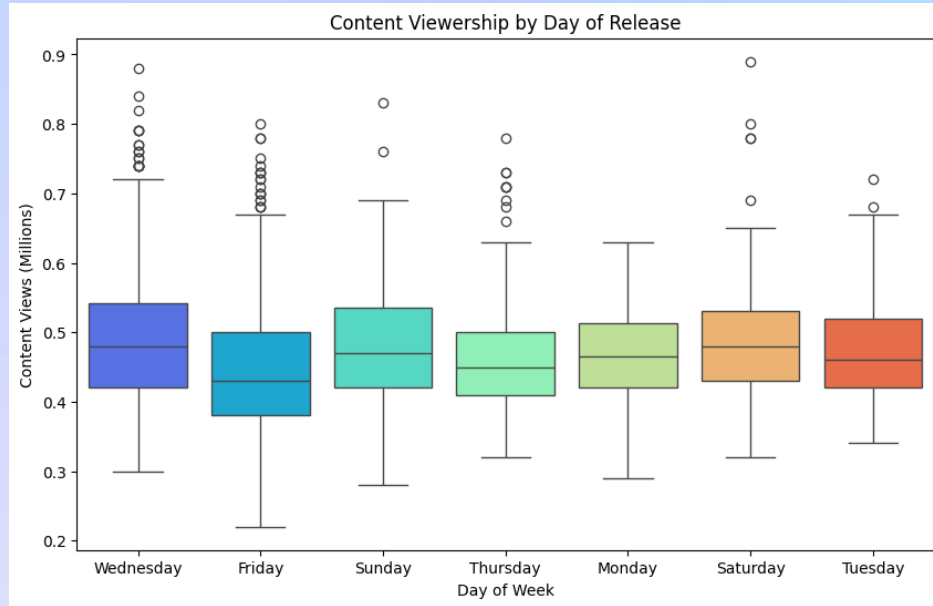


➤ Season distribution

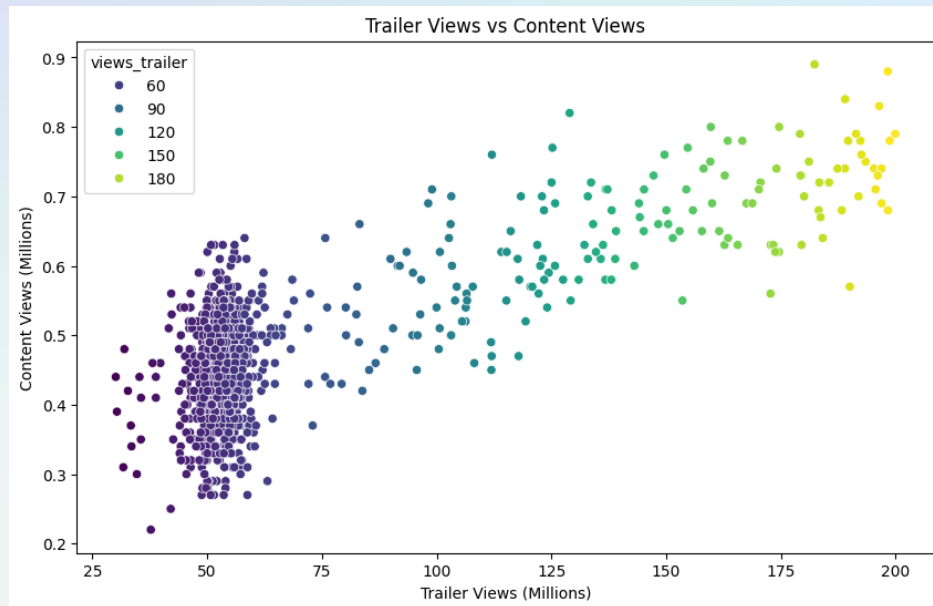


❖ Bivariate Analysis

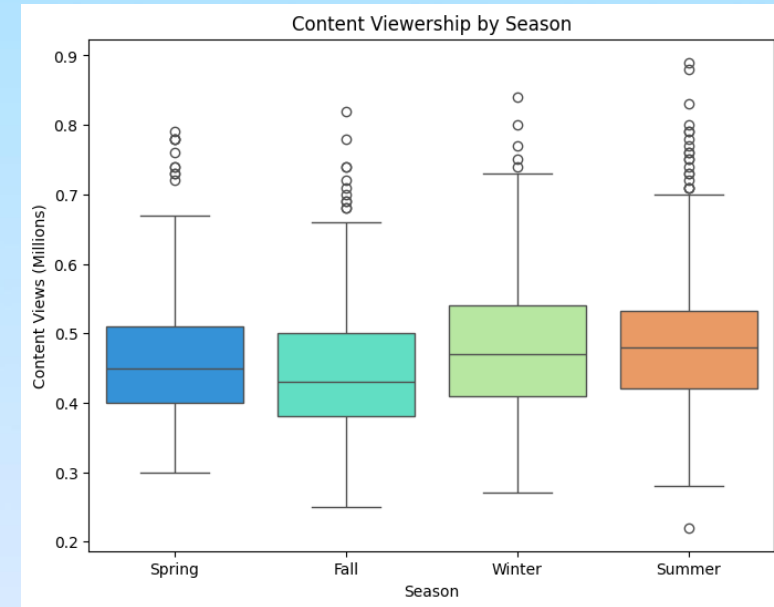
➤ Viewership vs Day of release



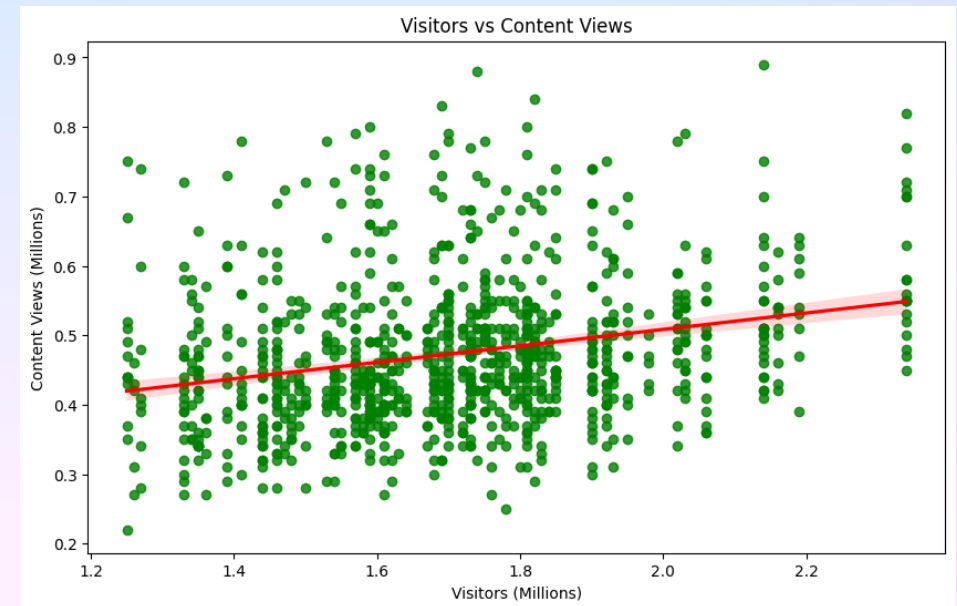
➤ Trailer views vs Content views



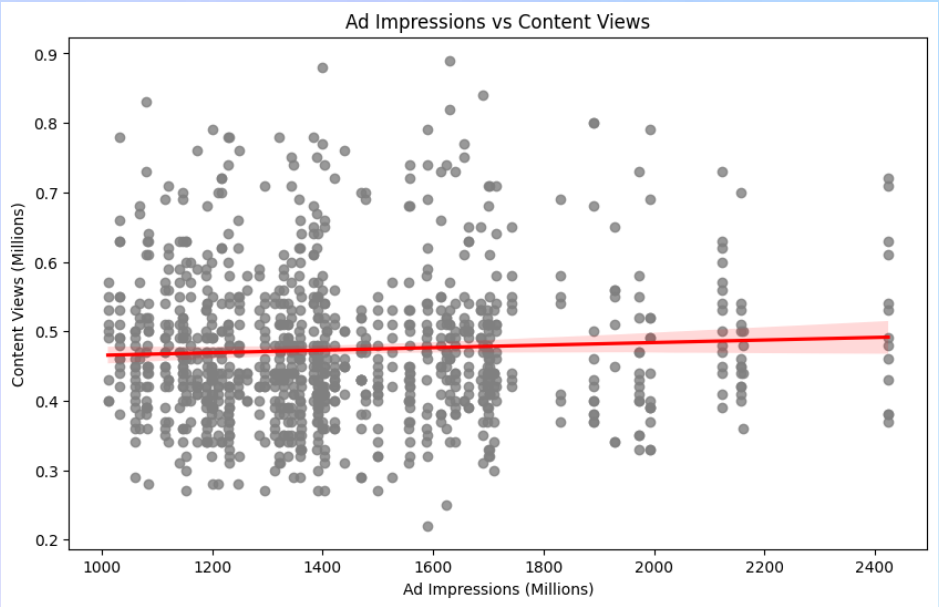
➤ Viewership vs Season



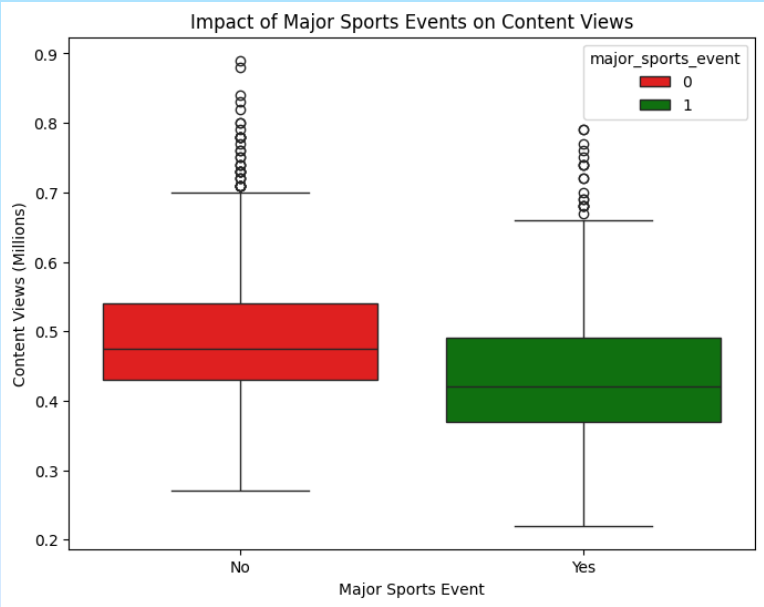
➤ Visitors vs Content views



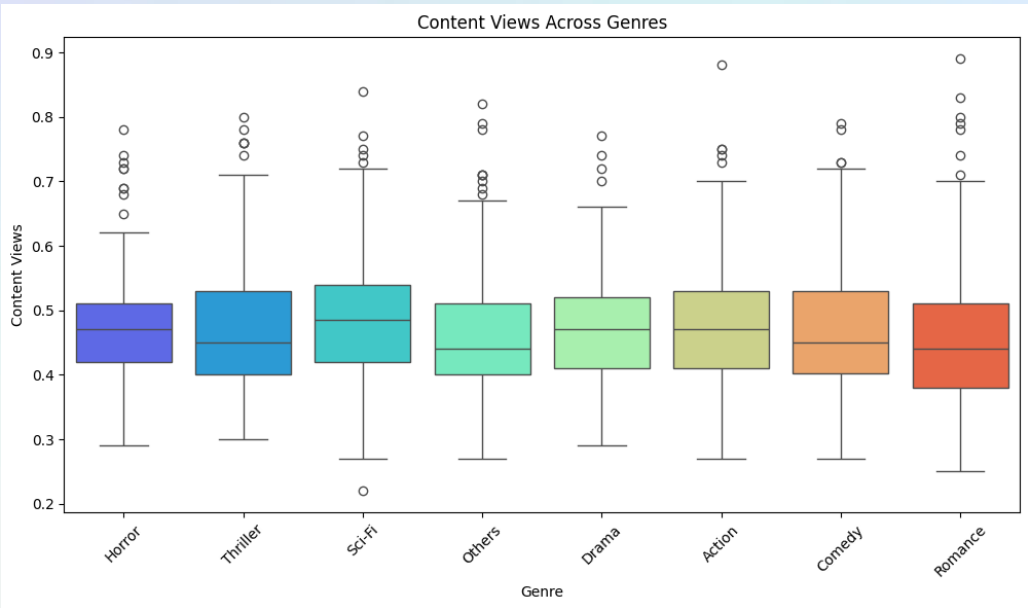
➤ AD impressions vs Content views



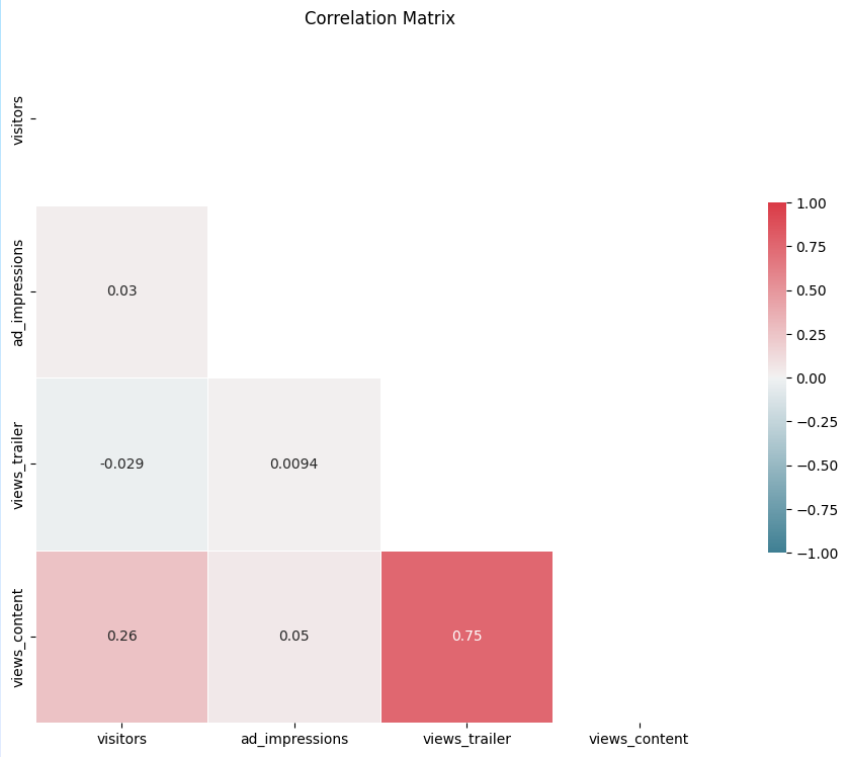
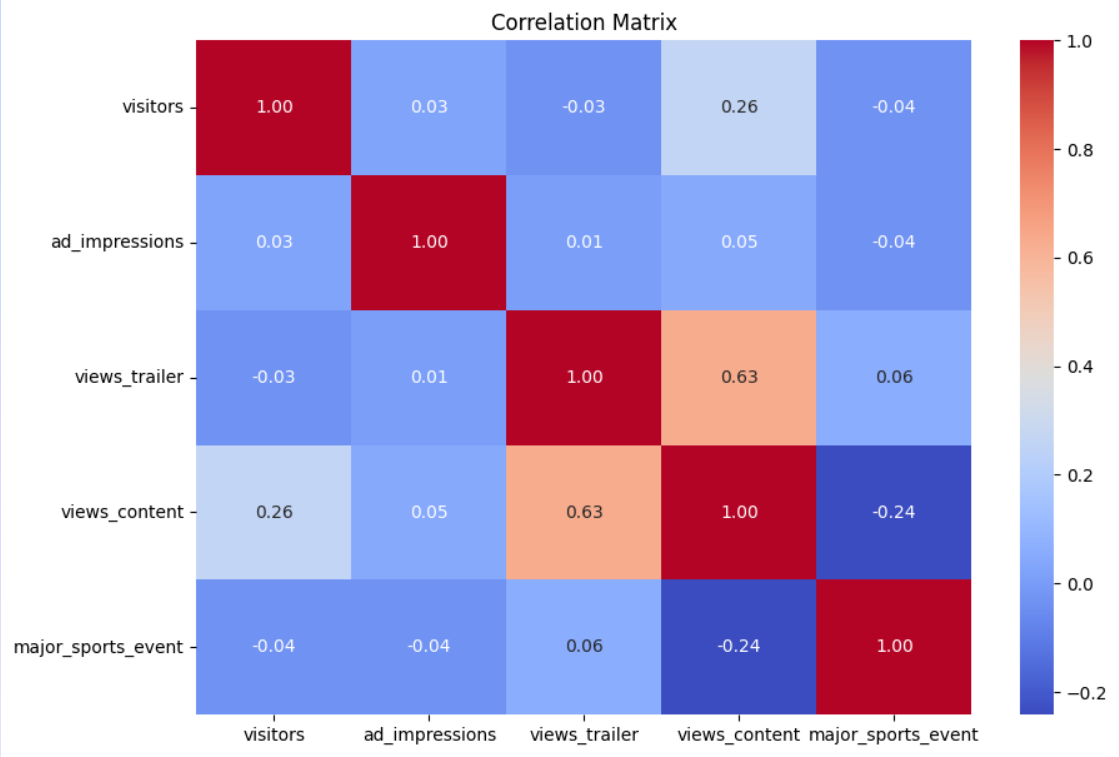
➤ Major sports event impact



➤ Genre vs Content views



❖ Correlation Analysis



❖ Answers to key questions

1. What does the distribution of content views look like?
 - The distribution shows that most contents receive moderate first-day viewership.
2. What does the distribution of genres look like?
 - The platform contains multiple genres with some genres dominating the release schedule.
3. The day of the week on which content is released generally plays a key role in the viewership. How does the viewership vary with the day of release?
 - Weekend releases tend to perform better.

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
0.467917	0.487826	0.494608	0.470619	0.446694	0.497955	0.484179

4. How does the viewership vary with the season of release?
 - Certain seasons show stronger audience engagement.

Fall	Spring	Summer	Winter
0.445357	0.467166	0.496803	0.484669

5. What is the correlation between trailer views and content views?
 - The correlation between trailer views and content views is 0.754. Hence it is strongly positive relationship.

❖ **Final Business Insights**

- Trailer views are the strongest driver of first-day content performance.
- Weekend releases tend to attract more audience engagement.
- Major sports events negatively impact OTT content viewership.
- Seasonal release timing influences audience behavior.
- Marketing impressions alone do not strongly guarantee success.
- Increasing platform visitors can improve overall viewership.
- Strategic release scheduling and strong trailer marketing can significantly improve content performance.

3. Model building - Linear Regression

➤ LINEAR REGRESSION MODEL SUMMARY

OLS Regression Results

=====

Dep. Variable:	views content	R-squared:	0.787
Model:	OLS	Adj. R-squared:	0.781
Method:	Least Squares	F-statistic:	143.8
Date:	Fri, 15 May 2026	Prob (F-statistic):	2.53e-245
Time:	17:39:32	Log-Likelihood:	1279.0
No. Observations:	800	AIC:	-2516.
Df Residuals:	779	BIC:	-2418.
Df Model:	20		
Covariance Type:	nonrobust		

MODEL PERFORMANCE METRICS

R-Squared	: 0.7743
RMSE	: 0.05
MAE	: 0.0399

	coef	std err	t	P> t	[0.025	0.975]

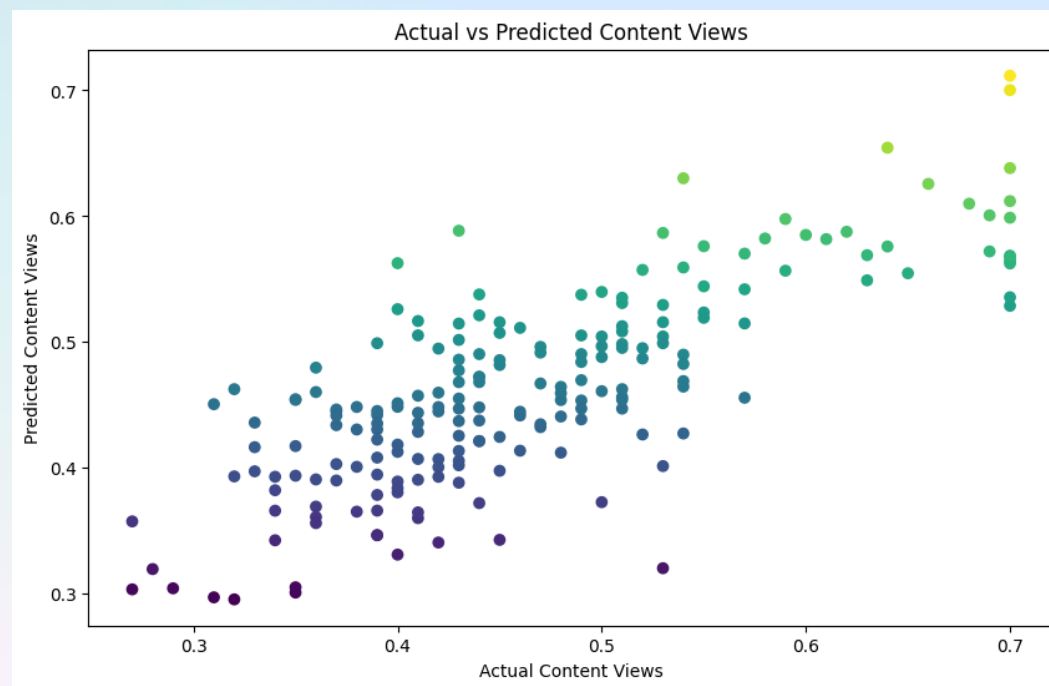
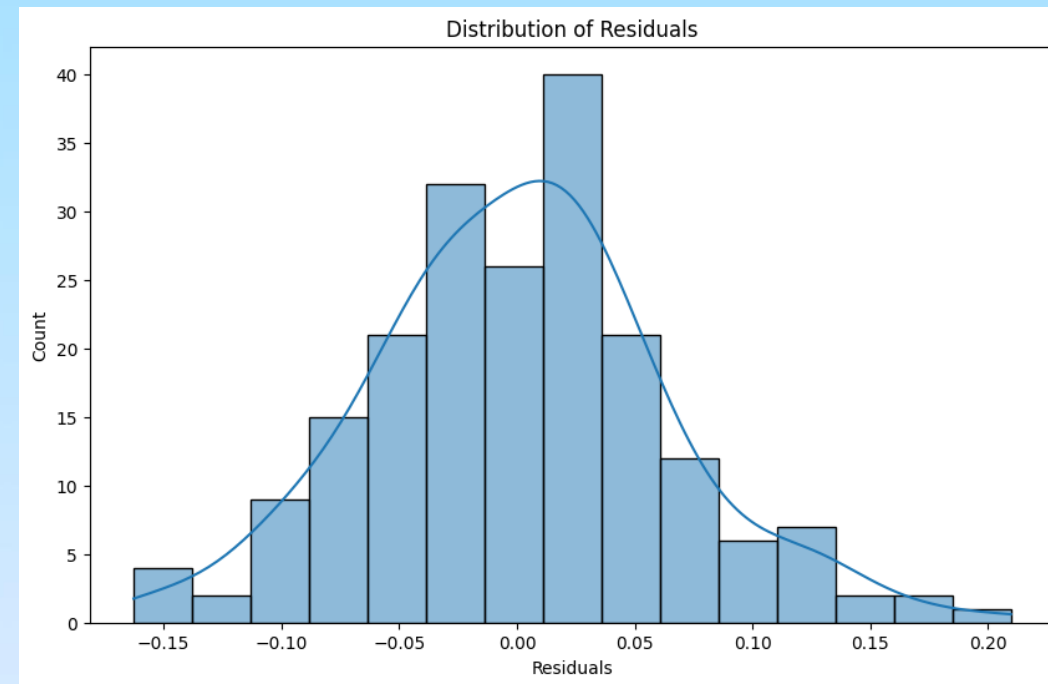
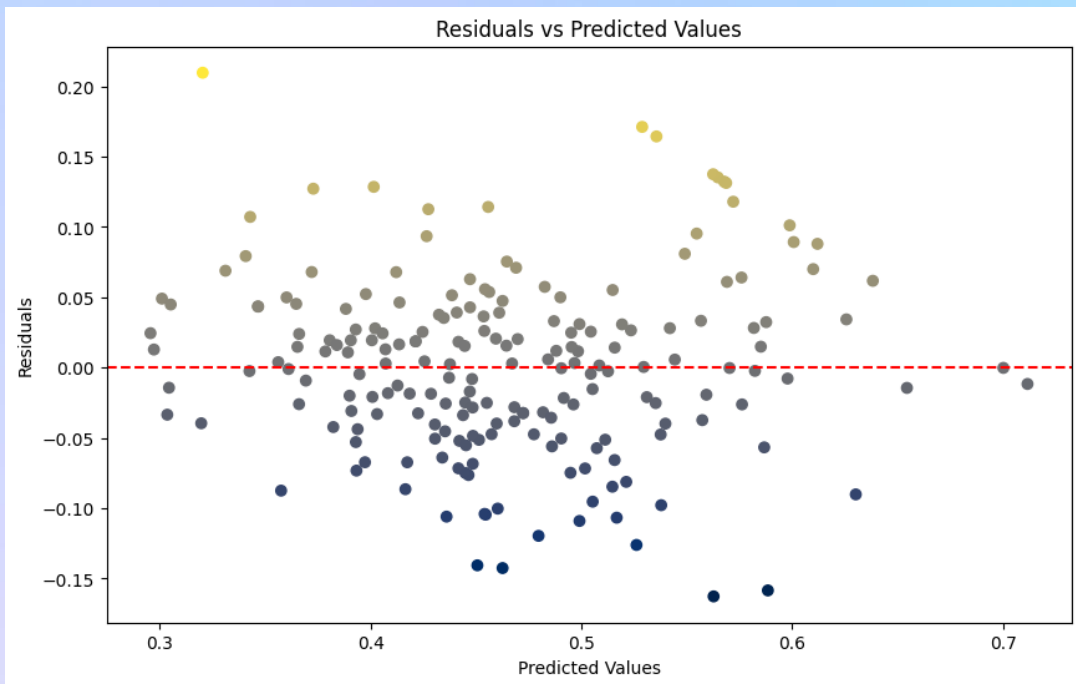
const	0.4750	0.002	271.021	0.000	0.472	0.478
visitors	0.0299	0.002	16.873	0.000	0.026	0.033
ad_impressions	0.0024	0.002	1.342	0.180	-0.001	0.006
major_sports_event	-0.0292	0.002	-16.280	0.000	-0.033	-0.026
views_trailer	0.0812	0.002	45.849	0.000	0.078	0.085
genre_Comedy	0.0014	0.002	0.563	0.573	-0.003	0.006
genre_Drama	0.0034	0.002	1.374	0.170	-0.001	0.008
genre_Horror	0.0030	0.002	1.215	0.225	-0.002	0.008
genre_Others	0.0022	0.003	0.742	0.459	-0.004	0.008
genre_Romance	-0.0004	0.002	-0.179	0.858	-0.005	0.004
genre_Sci-Fi	0.0030	0.002	1.242	0.215	-0.002	0.008
genre_Thriller	0.0036	0.002	1.479	0.140	-0.001	0.008
dayofweek_Monday	0.0067	0.002	3.712	0.000	0.003	0.010
dayofweek_Saturday	0.0144	0.002	7.692	0.000	0.011	0.018
dayofweek_Sunday	0.0098	0.002	5.287	0.000	0.006	0.013
dayofweek_Thursday	0.0060	0.002	3.181	0.002	0.002	0.010
dayofweek_Tuesday	0.0047	0.002	2.588	0.010	0.001	0.008
dayofweek_Wednesday	0.0233	0.002	11.706	0.000	0.019	0.027
season_Spring	0.0101	0.002	4.632	0.000	0.006	0.014
season_Summer	0.0191	0.002	8.705	0.000	0.015	0.023
season_Winter	0.0116	0.002	5.300	0.000	0.007	0.016
=====						
Omnibus:	2.150	Durbin-Watson:	2.038			
Prob(Omnibus):	0.341	Jarque-Bera (JB):	2.162			
Skew:	0.126	Prob(JB):	0.339			
Kurtosis:	2.965	Cond. No.	3.66			

- **Notes:** Standard Errors assume that the covariance matrix of the errors is correctly specified.

➤ **MODEL STATISTICS INTERPRETATION**

- 1. R-Squared: - Indicates how much variance in content views is explained by the independent variables. - Higher R² indicates better model performance.
- 2. RMSE: - Measures prediction error. - Lower RMSE indicates better prediction accuracy.
- 3. MAE: - Average absolute prediction error. - Lower MAE means more reliable predictions.
- 4. P-values: - Variables with p-value < 0.05 are statistically significant.
- 5. F-statistic: - Tests overall model significance. - Significant F-statistic indicates the model is useful.

<u>MODEL COEFFICIENTS WITH COLUMN NAMES</u>				<u>TOP IMPORTANT DRIVER VARIABLES</u>			
	Feature	Coefficient	P-Value		Feature	Coefficient	P-Value
0	const	0.474962	0.000000e+00	0	const	0.474962	0.000000e+00
4	views_trailer	0.081247	1.905697e-223	4	views_trailer	0.081247	1.905697e-223
1	visitors	0.029909	1.121568e-54	1	Visitors	0.029909	1.121568e-54
17	dayofweek_Wednesday	0.023313	2.859572e-29	17	dayofweek_Wednesday	0.023313	2.859572e-29
19	season_Summer	0.019118	1.872190e-17	19	season_Summer	0.019118	1.872190e-17
13	dayofweek_Saturday	0.014361	4.365047e-14	13	dayofweek_Saturday	0.014361	4.365047e-14
20	season_Winter	0.011602	1.506942e-07	20	season_Winter	0.011602	1.506942e-07
18	season_Spring	0.010063	4.236734e-06	18	season_Spring	0.010063	4.236734e-06
14	dayofweek_Sunday	0.009822	1.620440e-07	14	dayofweek_Sunday	0.009822	1.620440e-07
12	dayofweek_Monday	0.006683	2.198785e-04	12	dayofweek_Monday	0.006683	2.198785e-04
15	dayofweek_Thursday	0.006020	1.526799e-03	15	dayofweek_Thursday	0.006020	1.526799e-03
16	dayofweek_Tuesday	0.004674	9.841543e-03	16	dayofweek_Tuesday	0.004674	9.841543e-03
11	genre_Thriller	0.003622	1.396092e-01	3	major_sports_event	-0.029237	1.635459e-51
6	genre_Drama	0.003361	1.699553e-01				
7	genre_Horror	0.002981	2.247512e-01				
10	genre_Sci-Fi	0.002952	2.147279e-01				
2	ad_impressions	0.002371	1.798767e-01				
8	genre_Others	0.002154	4.585417e-01				
5	genre_Comedy	0.001359	5.733427e-01				
9	genre_Romance	-0.000438	8.582241e-01				
3	major_sports_event	-0.029237	1.635459e-51				



❖ **Final Business Insights**

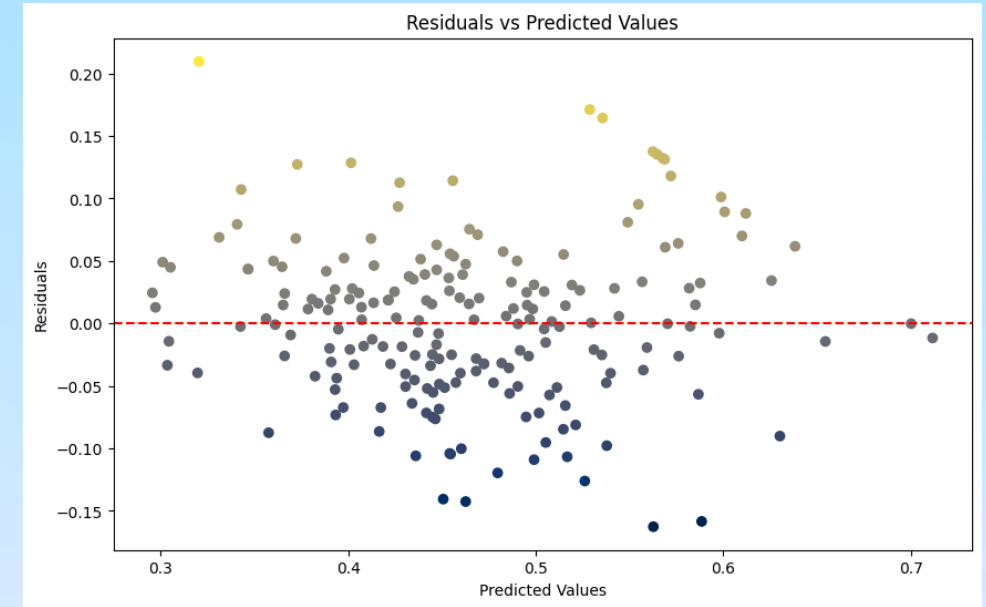
- Trailer views are expected to be one of the strongest positive drivers of first-day content viewership.
- Major sports events may negatively affect OTT traffic and reduce content engagement.
- Release timing (day and season) influences performance.
- Higher platform visitors generally improve viewership.
- Statistically significant variables should be prioritized for marketing and release strategy planning.
- The regression model helps ShowTime predict content performance and optimize business decisions.

4. Testing the assumptions of linear regression model

➤ Assumption 1: Linearity

Interpretation:

- Residuals should be randomly scattered around zero.
- No visible pattern indicates linearity assumption is satisfied.



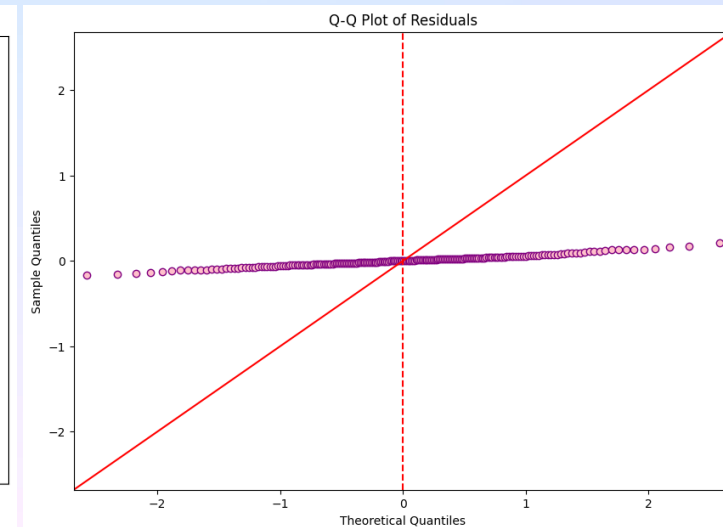
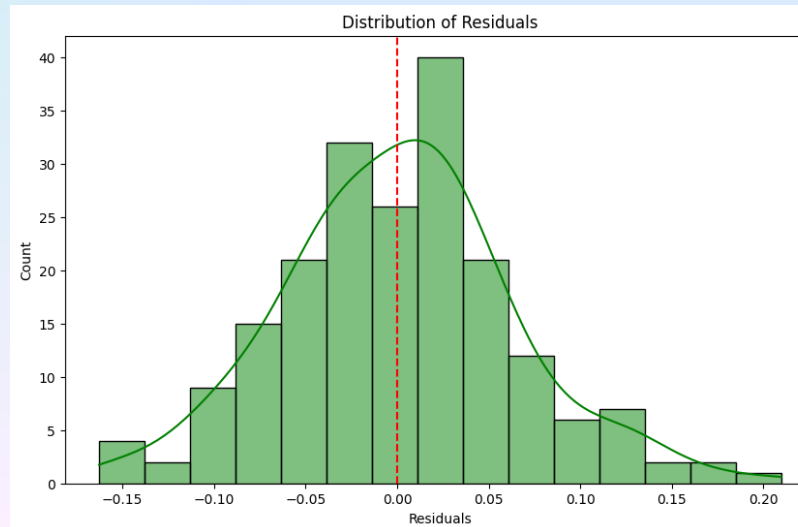
➤ Assumption 2: Normality of Residuals

Shapiro-Wilk Test Statistic: 0.9922

P-value: 0.3645

Conclusion:

- Residuals are normally distributed.
- Normality assumption is satisfied.



➤ Assumption 3: Homoscedasticity

- 'Lagrange Multiplier Statistic': np.float64(21.916613200974467),
- 'P-value': np.float64(0.34505253543346714),
- 'F-statistic': np.float64(1.1014710112745607),
- 'F-test P-value': np.float64(0.3514604347479038)

Conclusion:

- Homoscedasticity assumption is satisfied.
- Variance of residuals is constant.

➤ Assumption 4: Multicollinearity

Interpretation: -

- $VIF < 5 \rightarrow$ No multicollinearity concern
- $VIF \text{ 5-10} \rightarrow$ Moderate multicollinearity
- $VIF > 10 \rightarrow$ Serious multicollinearity issue

VIF			VIF		
Feature	VIF	Feature	VIF	Feature	VIF
0 const	1.000000	6 genre_Drama	1.949123	11 genre_Thriller	1.952752
1 Visitors	1.022671	7 genre_Horror	1.957056	12 dayofweek_Monday	1.054613
2 ad_impressions	1.016695	8 genre_Others	2.744608	13 dayofweek_Saturday	1.134732
3 major_sports_event	1.051825	9 genre_Romance	1.951451	14 dayofweek_Sunday	1.123843
4 views_trailer	1.023142	10 genre_Sci-Fi	1.841998	15 dayofweek_Thursday	1.166108
5 genre_Comedy	1.893001			16 dayofweek_Tuesday	1.062378
				17 dayofweek_Wednesday	1.291416
				18 season_Spring	1.536915
				19 season_Summer	1.570450
				20 season_Winter	1.559595

➤ Assumption 5: Independence of errors

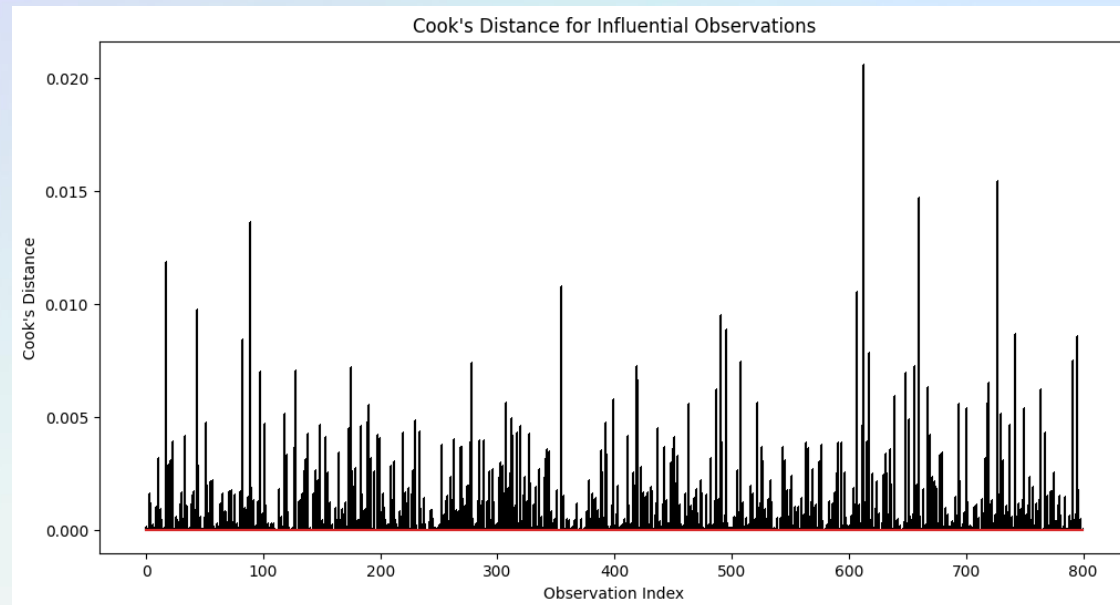
Durbin-Watson Statistic: 1.8147

Conclusion:

- Residuals are independent.
- No autocorrelation detected.

➤ Assumption 6: Outlier influence

- Cook's Distance Threshold: 0.005
- Number of Influential Points: 43
- Interpretation: - Very few influential points indicate stable model performance.



❖ Final Assumption Test Summary

- Linearity: Residual plots should show random distribution.
 - Normality: Residuals should approximately follow normal distribution.
 - Homoscedasticity: Residual variance should remain constant.
 - Multicollinearity: VIF values should ideally remain below 5.
 - Independence: Durbin-Watson statistic should be close to 2.
 - Influential Outliers: Cook's distance identifies influential observations.
-
- If all assumptions are reasonably satisfied, the linear regression model is considered reliable.

5. Model performance evaluation

<u>LINEAR REGRESSION MODEL PERFORMANCE</u>			
	Metric	Training Data	Testing Data
0	R-Squared	0.7868	0.7743
1	RMSE	0.0489	0.0500
2	MAE	0.0389	0.0399
3	MAPE	0.0861	0.0908

➤ PERFORMANCE METRIC INTERPRETATION

1. R-Squared

- Measures how much variance in the target variable is explained by the model.
- Higher R^2 indicates better model fit.

2. RMSE (Root Mean Squared Error)

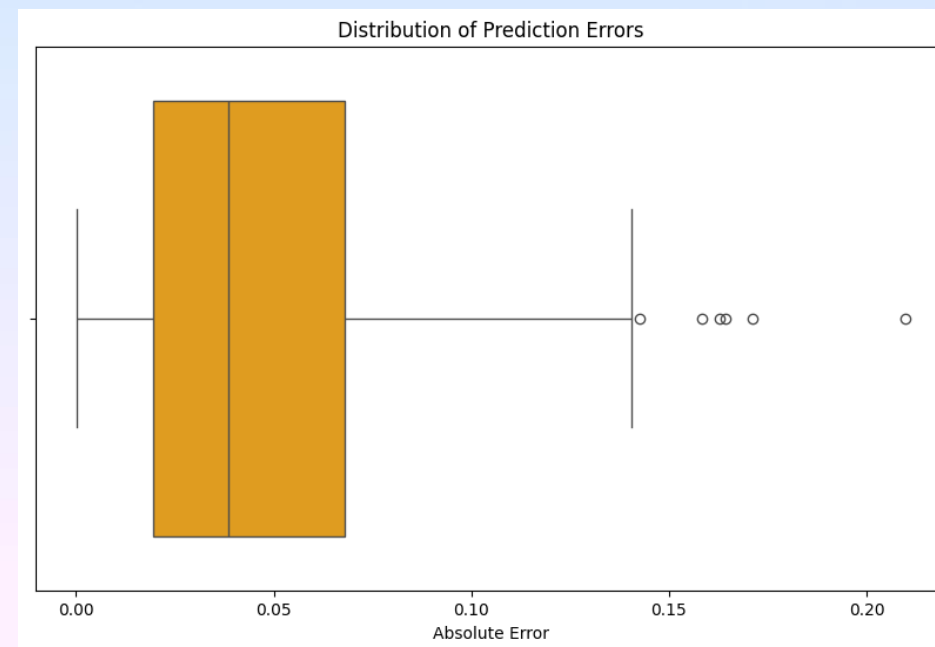
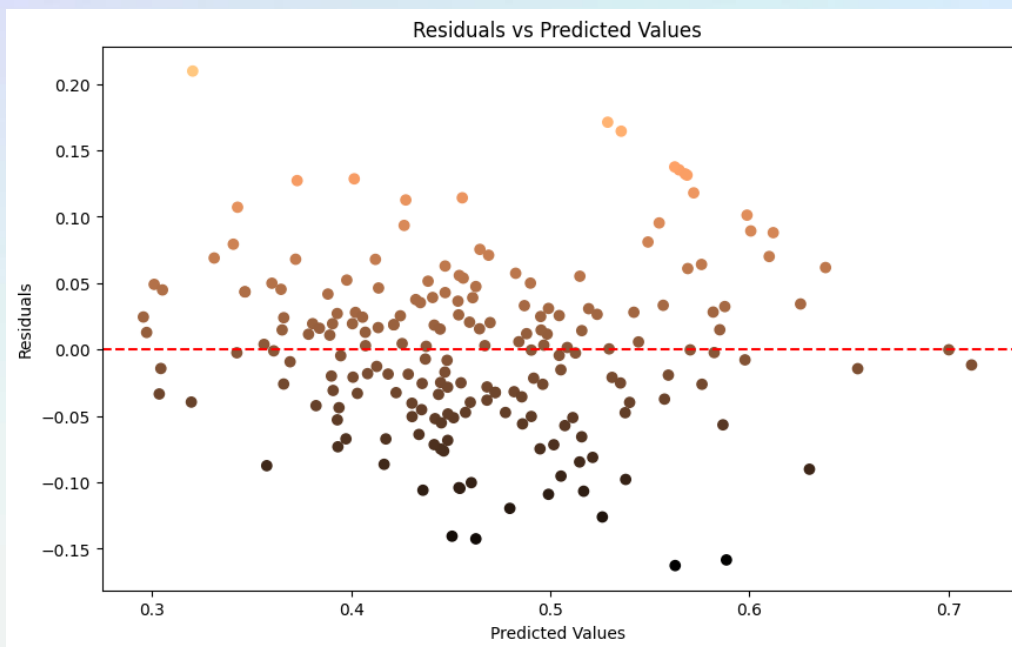
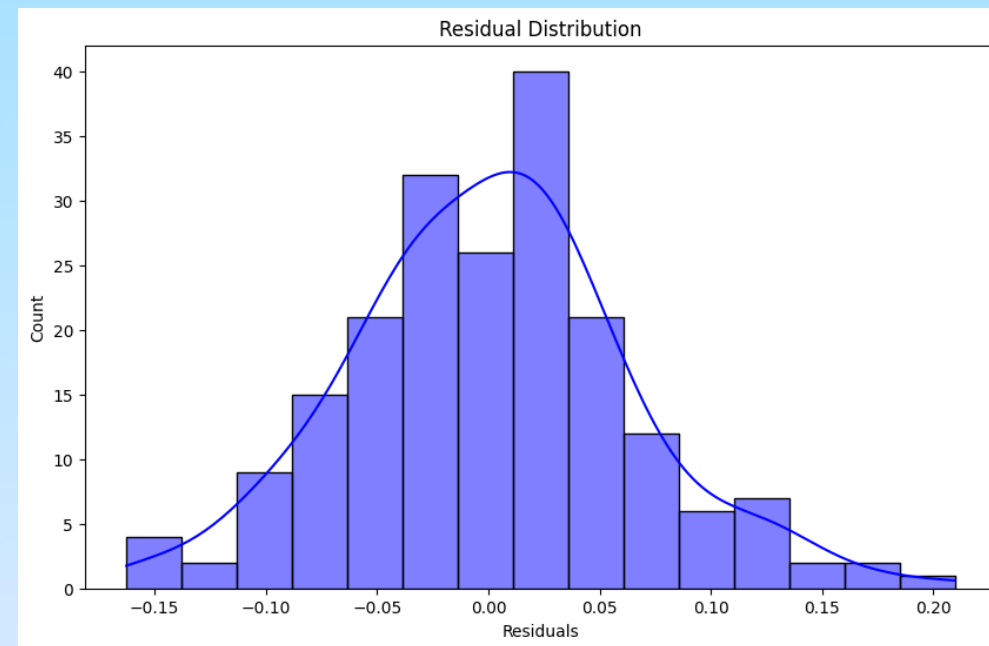
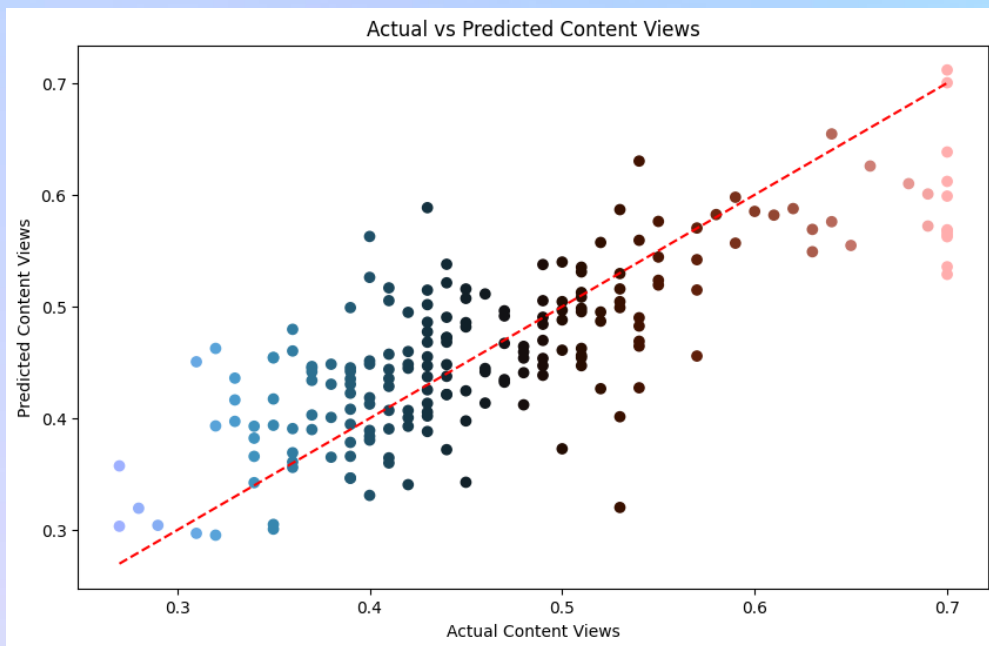
- Measures average prediction error.
- Lower RMSE indicates better prediction accuracy.

3. MAE (Mean Absolute Error)

- Average absolute difference between actual and predicted values.
- Lower MAE means more reliable predictions.

4. MAPE (Mean Absolute Percentage Error)

- Measures prediction error percentage.
- Lower MAPE indicates better forecasting performance.



TOP MODEL DRIVER VARIABLES		
	Feature	Coefficient
0	const	0.472112
4	views_trailer	0.063211
1	visitors	0.029013
17	dayofweek_Wednesday	0.020905
19	season_Summer	0.017281
13	dayofweek_Saturday	0.012379
20	season_Winter	0.012181
18	season_Spring	0.009842
14	dayofweek_Sunday	0.009728
15	dayofweek_Thursday	0.006537

➤ Business Interpretation

- The model successfully explains a significant portion of variation in first-day OTT content viewership.
- Trailer views are likely to be one of the strongest predictors of content performance.
- Release timing and seasonal factors influence viewership.
- The model shows acceptable prediction accuracy based on:
 - R-Squared
 - RMSE
 - MAE
 - MAPE
- Small difference between training and testing metrics indicates the model generalizes well and is not overfitting.
- The model can help ShowTime optimize:
 - Marketing strategy
 - Trailer promotions
 - Release scheduling
 - Audience engagement planning

❖ Final Conclusion

- The Linear Regression model performs reasonably well for predicting first-day OTT content viewership.
- The model demonstrates:
 - Good explanatory power
 - Stable prediction capability
 - Reliable error performance
- The analysis identifies important business drivers that influence OTT content success.

6. Actionable Insights & Recommendations

- Trailer Views: Highly Significant Positive Predictor
- Platform Visitors: Significant Positive Predictor
- Major Sports Events: Significant Negative Predictor
- Day of Release: Important Predictor
- Season of Release: Moderately Significant Predictor
- Advertisement Impressions: Weak Predictor

❖ Overall Business Insights

- **Audience Anticipation Drives Success**
 - Trailer engagement is the strongest indicator of content popularity.
 - **Business Focus:** Build excitement before release.
- **Timing Is a Critical Success Factor**
 - Release scheduling significantly affects first-day performance.
 - **Business Focus:** Optimize launch calendars carefully.
- **Customer Engagement Must Increase**
 - Higher platform traffic improves overall content consumption.
 - **Business Focus:** Increase active users and retention.
- **External Events Affect OTT Consumption**
 - Sports events negatively impact entertainment consumption patterns.
 - **Business Focus:** Avoid competitive entertainment timing.

❖ Final Conclusion

The Linear Regression model successfully identified the major drivers influencing first-day OTT content viewership.

- The most important factors are:
 - Trailer popularity
 - Platform visitors
 - Release timing
 - Seasonal trends
 - Sports-event competition
- These insights can help ShowTime:
 - Improve marketing efficiency
 - Optimize release schedules
 - Increase audience engagement
 - Predict content performance more accurately
 - Maximize OTT platform growth and profitability